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LAB#08
Azmay noor**Basic Fundamentals of Web Development**

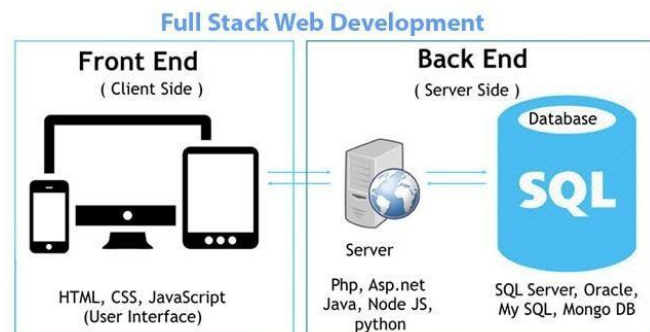
OBJECTIVE: To Implement and Understand Introduction to web development along with web development concepts using HTML.

THEORY:**Introduction to Web Development**

It is the process of building, creating and maintaining of websites. It involves coding and using programming languages to create websites that are functional, userfriendly, and perform well.

It includes aspects such as;

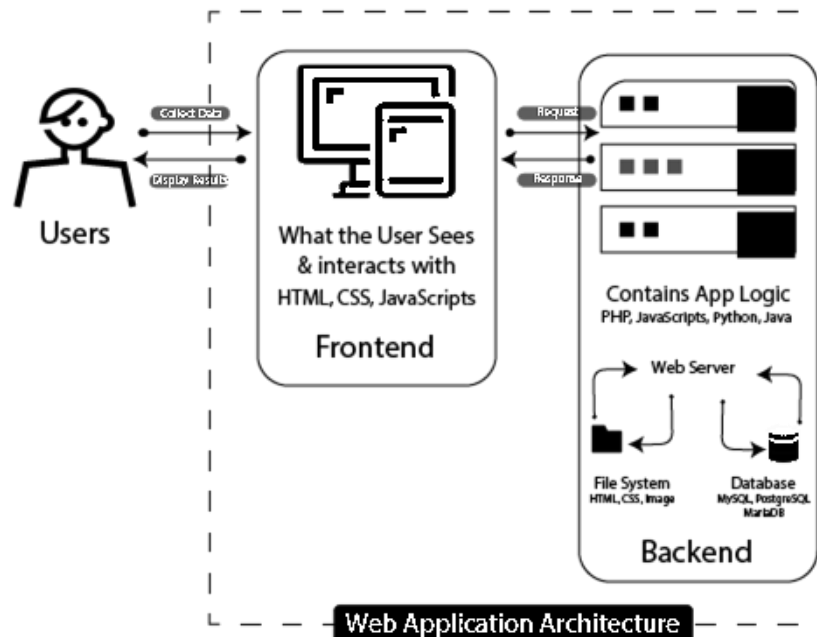
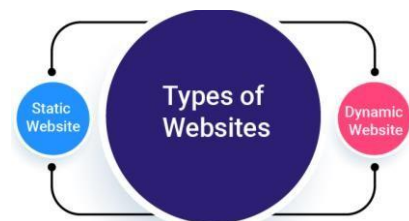
- Web designing
- Web Programming
- Web Publishing
- Database Management



It is the creation of an application that work over the internet. The main components of web development can be categorized as;

- **Front-end:** The part of the website that users see and interact with (HTML, CSS, JavaScript) also known as client-side development.
- **Back-end:** Handles server-side logic, databases, and data processing that users don't see directly. Common languages are Python, Java, Node.js, etc.

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Introduction to Static & Dynamic Website

- **Full stack development:** Involves both frontend and backend work. A fullstack developer builds both the visible & the server parts of a website or app.

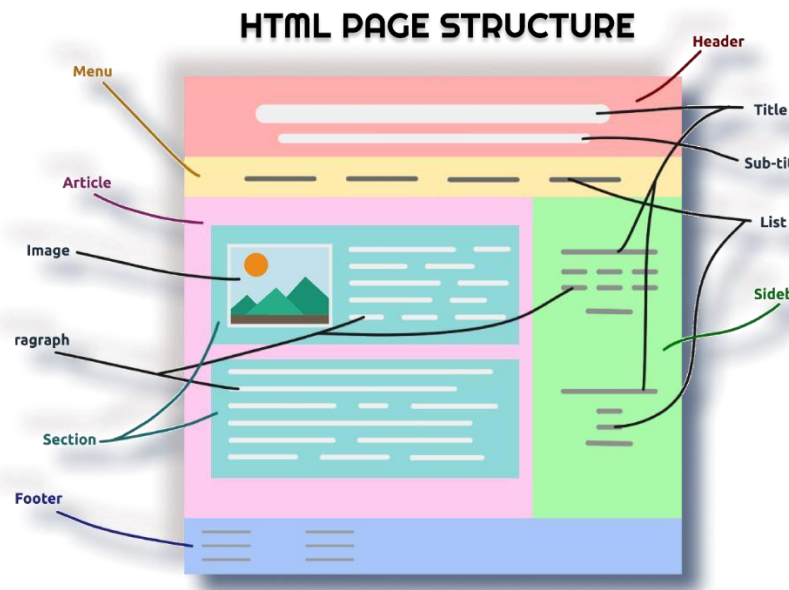
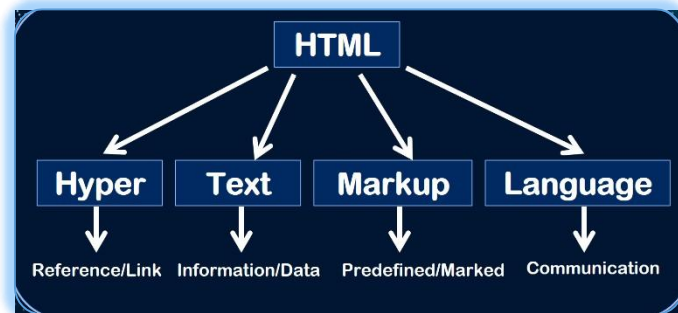


- **Static Website:** Contains fixed content that doesn't change without modifying the code. Good for simple sites with limited interactivity.
- **Dynamic Website:** Content can change in response to user interactions or server-side data. Requires backend logic and database support.

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Introduction to HTML

HTML (HyperText Markup Language) is the standard language for creating web pages. It defines the structure of a webpage using tags that organize content into headings, paragraphs, links, images, etc. HTML elements tell the browser how to display the content.

**Example Code**

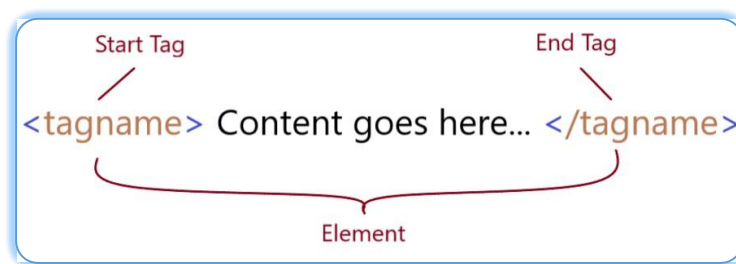
```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
```

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```
</body>
</html>
```

Example Explained

- The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- The **<html>** element is the root element of an HTML page
- The **<head>** element contains meta information about the HTML page
- The **<title>** element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The **<body>** element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The **<h1>** element defines a large heading
- The **<p>** element defines a paragraph

HTML Tags and Defined Structure

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	<i>none</i>	<i>none</i>

HTML Tags

<u>Tags</u>	<u>Purpose</u>	<u>Syntax</u>
Title Tag	It is used to specify the title of a web page, which is typically displayed in the browser's title bar.	<title> Page Title </title>

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Heading Tags	It is used to display headings h1, h2, h3, h4, h5, h6	<h1> Hello World </h1>
Paragraph Tag	It is used to add paragraphs.	<p> A Paragraph</p>
Br Tag	It is used to add next line (line breaks) to our page.	

Bold, Italic & Underline Tags	These Tags are used to highlight text to our page.	SSUET <i>SSUET</i> <u>SSUET</u>
Center Tag	It is used for centering content within a web page.	<center> Content</center>
Subscript & Superscript Tag	They are used to display a horizontal ruler, used to separate content.	The <sub> subscripted </sub> The <sup> superscripted </sup>
Button Tag	It is used to create a clickable button in HTML. It can be used to perform certain actions i.e. submit, reset etc.	<button>Click Me!</button>

HTML Attributes and Defined Structure

Attributes are used to add more information to the tag. All HTML elements can have attributes. Some Attributes are **bgcolor href, src, width, height** etc.

Image Tag	Image Tag is used to add images to our Html page.	
Iframe Tag	It is used to embed content from other websites i.e. YouTube, a Google Map etc.	<iframe src = "URL"width = "pixels" height = "pixels"> </iframe>
Video Tag	It is used to add video	<video src="myVid.mp4">My Video </video>
Anchor Tag	Anchor Tag is used to add Links to our Html page.	 Google

HTML Lists

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Lists are used to represent real life list data.

**Example Output**

An unordered HTML list:

- Item
- Item
- Item
- Item

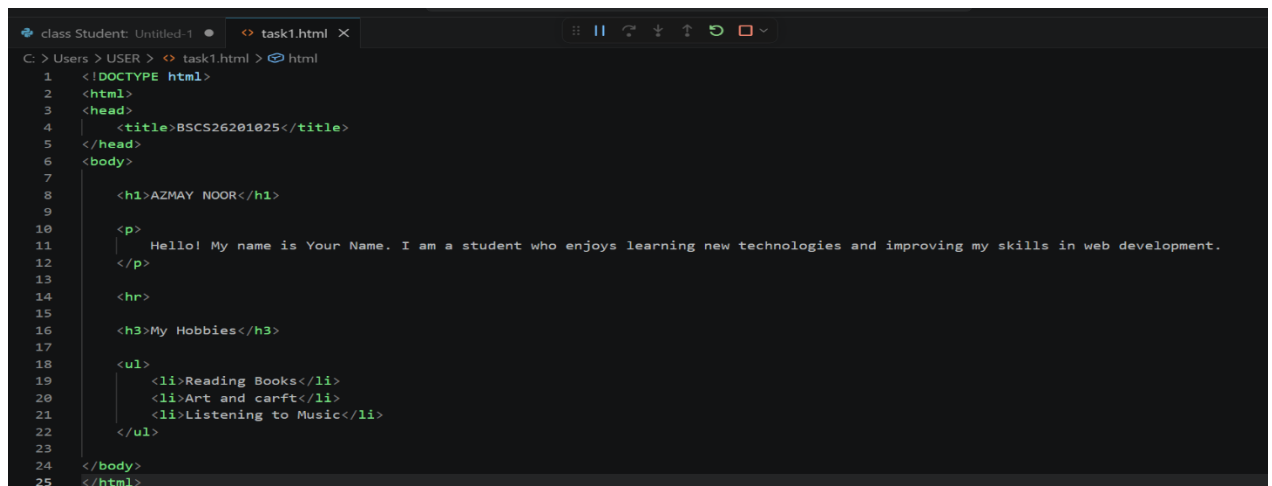
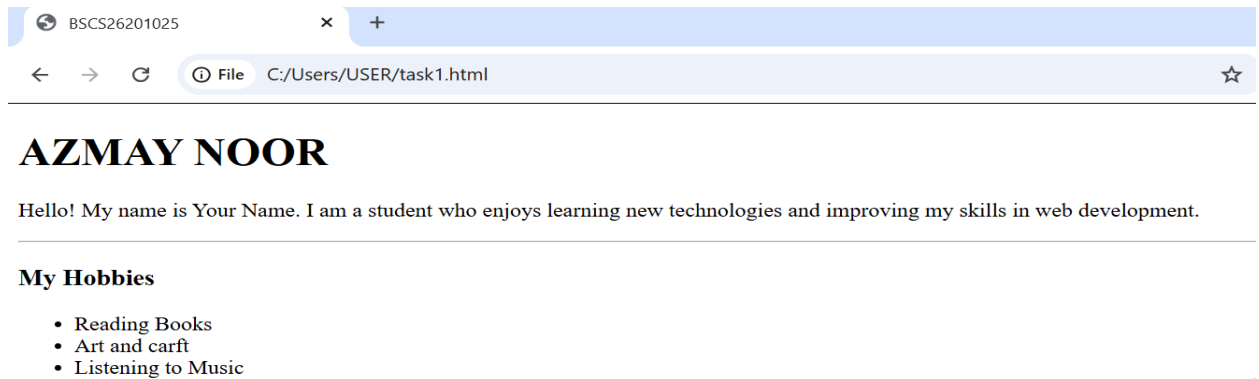
An ordered HTML list:

1. First item
2. Second item
3. Third item
4. Fourth item

LAB ACTIVITY # 1

Create a Basic HTML Page Structure

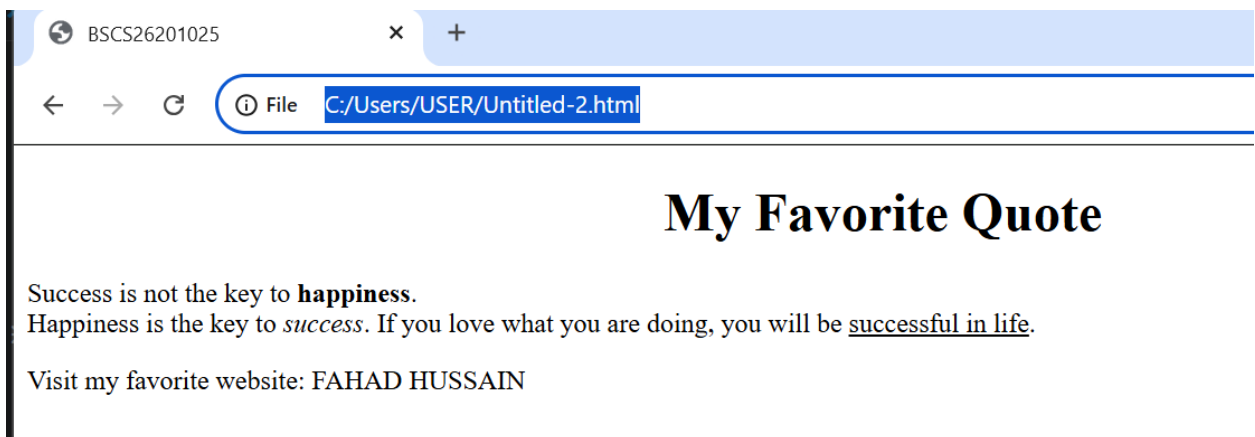
1. Create a new file named **task1.html**.
2. Write a complete HTML document structure using:
 - `<!DOCTYPE html>`
 - `<html>`, `<head>`, `<body>`
 - A `<title>` of your **Registration Number**
3. Inside the `<body>`, add:
 - A main heading (`<h1>`) with your name
 - A paragraph (`<p>`) introducing yourself
 - A horizontal rule (`<hr>`)
 - A smaller heading (`<h3>`) saying "My Hobbies"
 - An unordered list (`<ul>`) of at least 3 hobbies



Lab Activity#2

Practice Text Formatting & Links

1. Create a new file named **task2.html**.
2. Write a complete HTML document structure using:
 - `<!DOCTYPE html>`
 - `<html>`, `<head>`, `<body>`
 - A `<title>` of your **Registration Number**
3. Inside the `<body>`, add:
 - A centered heading (`<center>` or CSS style) saying "My Favorite Quote"
 - A paragraph with a quote, using:
 - Bold (`<b>`) for important words
 - Italic (`<i>`) for emphasis
 - Underline (`<u>`) for one phrase
 - A line break (`<br>`) between two parts of the quote
 - A hyperlink (`<a>`) to a website you like (e.g., YouTube, Wikipedia)




```
class Student:
    def __init__(self, name, age, gender):
        self.name = name
        self.age = age
        self.gender = gender

    def display(self):
        print(f"Name: {self.name}, Age: {self.age}, Gender: {self.gender}")

# Create an object of the Student class
student = Student("FAHAD", 20, "Male")

# Call the display method
student.display()
```

task1.html

Untitled-2.html X

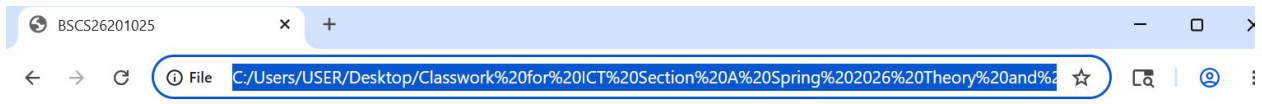
C: > Users > USER > <> Untitled-2.html > html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>BSCS26201025</title>
5  </head>
6  <body>
7
8      <center>
9          <h1>My Favorite Quote</h1>
10     </center>
11
12     <p>
13         Success is not the key to <b>happiness</b>.<br>
14         Happiness is the key to <i>success</i>. If you love what you are doing,
15         you will be <u>successful in life</u>.
16     </p>
17
18     <p>
19         Visit my favorite website:
20         <a href="https://youtu.be/ItuEZgCQo18?si=orYeCNCyBTZ4Nt3S" target="_blank">FAHAD HUSSAIN</a>
21     </p>
22
23 </body>
24 </html>
```

LAB ACTIVITY #3

Work with Images & Attributes

1. Create a new file named **task3.html**.
2. Write a complete HTML document structure using:
 - `<!DOCTYPE html>`
 - `<html>`, `<head>`, `<body>`
 - A `<title>` of your **Registration Number**
3. Inside the **<body>**, add:
 - A **heading**: "My Favorite Place"
 - An image (****) from the internet (use `src`, `width`, `height` attributes)
 - A caption below the image using **<p>**
 - A link to Google Maps (**<a>**) with the text "View on Map"
 - Use the **target="_blank"** attribute to open the link in a new tab



My Favorite Place



Mazar-e-Quaid is one of my favorite places in Karachi. It is the mausoleum of Quaid-e-Azam Muhammad Ali Jinnah and is a famous national monument in Pakistan.

[View on Map](#)

```
File Edit Selection View Go Run Help Search
class Student: Untitled-1 task1.html task3.html X Release Notes: 1.123.0
C:\Users\USER\Desktop>Classwork for ICT Section A Spring 2026 Theory and Lab Section A - Classroom_files > task3.html > ...
1 <!DOCTYPE html>
2
3 <html>
4 <head>
5   <title>BSCS26201025</title>
6 </head>
7 <body>
8   ...
9
10 <h1>My Favorite Place</h1>
11
12 
16
17 <p>
18   Mazar-e-Quaid is one of my favorite places in Karachi. It is the mausoleum of Quaid-e-Azam Muhammad Ali Jinnah and is a famous national
19 </p>
20
21 <a href="https://maps.google.com/?q=Mazar-e-Quaid,Karachi" target="_blank">
22   View on Map
23 </a>
24   ...
25
26 </body>
27 </html>
```

Lab Activity#4

Create a Simple Contact Section

1. Create a new file named **task4.html**.
2. Write a complete HTML document structure using:
 - `<!DOCTYPE html>`
 - `<html>`, `<head>`, `<body>`
 - A `<title>` of your **Registration Number**
3. Inside the `<body>`, Tag build a **contact section** that includes: ☐ A **heading**: "Contact Me" ☐ An **unordered list** with:
 - Email (as a mailto link)
 - Phone number (plain text)
 - LinkedIn profile (hyperlink)
- A **"Send Message" button** (`<button>`) (no functionality needed)



Contact Me

- Email: azmay@example.com
- Phone: +92 300 1234567
- LinkedIn: [Visit My LinkedIn Profile](https://www.linkedin.com/)

Send Message

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>BSCSZ26201025</title>
5 </head>
6 <body>
7
8   <h1>Contact Me</h1>
9
10  <ul>
11    <li>
12      Email:
13      <a href="mailto:azmay@example.com">
14        azmay@example.com
15      </a>
16    </li>
17
18    <li>
19      Phone: +92 300 1234567
20    </li>
21
22    <li>
23      LinkedIn:
24      <a href="https://www.linkedin.com/" target="_blank">
25        Visit My LinkedIn Profile
26      </a>
27    </li>
28  </ul>
29
```

```

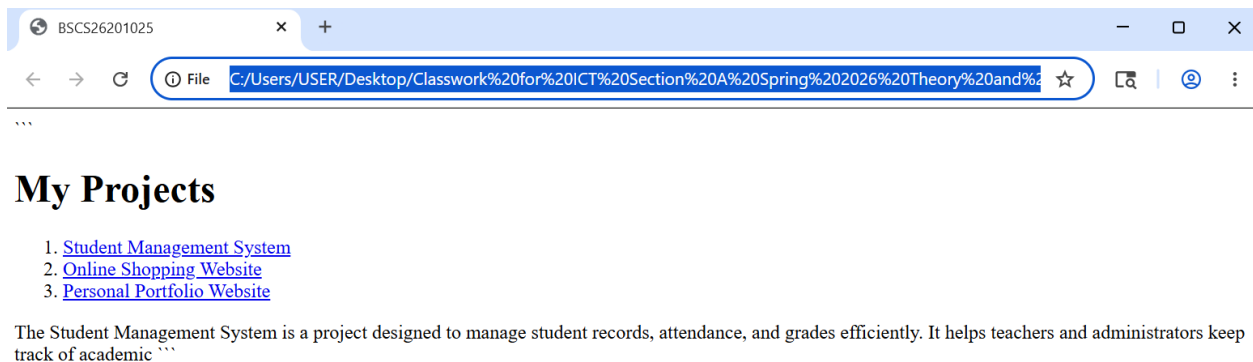
10      <ul>
11      |   </li>
12      |
13      |   </ul>
14      |
15      |   <button>Send Message</button>
16      |
17      |   </body>
18      |
19      |   </html>
20      |
21      |
22      |
23      |
24      |
25      |
26      |
27      |
28      |
29      |
30      |
31      |
32      |
33      |

```

LAB ACTIVITY # 5

Build a Mini Project List

1. Create a new file named **task5.html**.
2. Write a complete HTML document structure using:
 - `<!DOCTYPE html>`
 - `<html>`, `<head>`, `<body>`
 - A `<title>` of your **Registration Number**
3. Inside the **<body>**, Tag make a **"My Projects"** section with::
 - A **heading**
 - An ordered list (****) of 3 project names
 - Each project name should be a link (**<a>**) to a placeholder like #
 - Below the list, add a short description of one project using **<p>**



```
class Student:
    def __init__(self, name, age, grade):
        self.name = name
        self.age = age
        self.grade = grade

    def display_info(self):
        print(f"Student Name: {self.name}, Age: {self.age}, Grade: {self.grade}")

# Create a student object
student = Student("John Doe", 15, "10th Grade")

# Display student information
student.display_info()
```

C: > Users > USER > Desktop > Classwork for ICT Section A Spring 2026 Theory and Lab Section A - Classroom_files > > Untitled-2.f

```
1  <!DOCTYPE html>
2
3  <html>
4  <head>
5      <title>BSCS26201025</title>
6  </head>
7  <body>
8
9      ``
10 <h1>My Projects</h1>
11
12 <ol>
13     <li><a href="#">Student Management System</a></li>
14     <li><a href="#">Online Shopping Website</a></li>
15     <li><a href="#">Personal Portfolio Website</a></li>
16 </ol>
17
18 <p>
19     The Student Management System is a project designed to manage student records,
20     attendance, and grades efficiently. It helps teachers and administrators keep
21     track of academic
22     ``
23
```

Lab Assignment

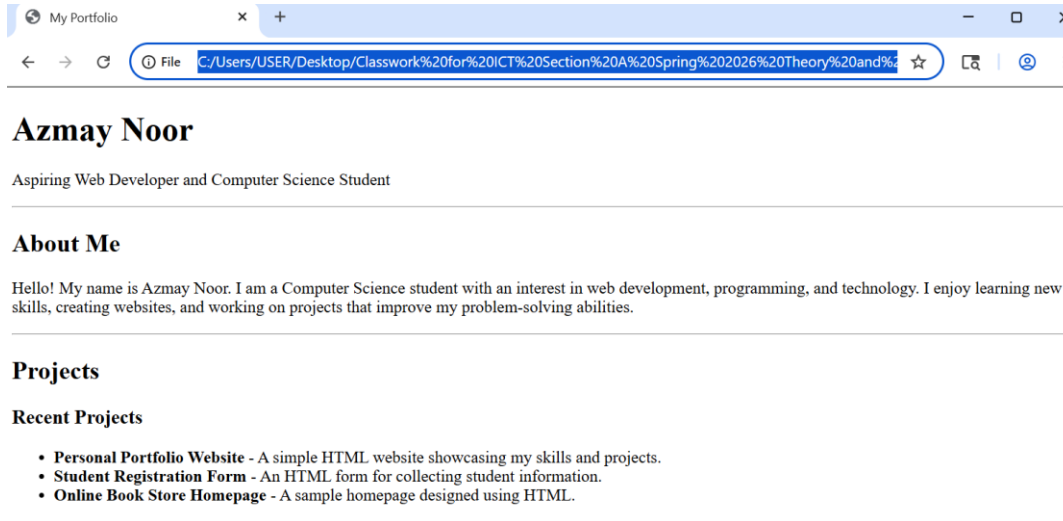
Create a personal portfolio website using only **HTML**. Your portfolio should include the following sections:

1. **Header:** Include your name and a short tagline or description about yourself.
2. **About Me Section:** Write a brief paragraph introducing yourself, your background, or your interests.
3. **Projects Section:** List at least 3 of your recent or favorite projects (you can create placeholders if you don't have real projects).
4. **Contact Section:** Provide contact details, such as your email address, or a simple message like "Get in touch!"

Requirements:

- Use semantic HTML tags (e.g., <header>, <section>, <footer>).

- Ensure proper use of headings (<h1> to <h3>).
- Use lists (or) where appropriate.
- Add at least one hyperlink (e.g., a link to a LinkedIn profile, GitHub or a placeholder email link).



```

52 <!-- Contact Section -->
53 <section>
54   <h2>Contact</h2>
55   <p>Get in touch!</p>
56
57   <ul>
58     <li>Email: <a href="mailto:azmay@example.com">azmay@example.com</a></li>
59     <li>GitHub: <a href="https://github.com" target="_blank">Visit My GitHub</a></li>
60   </ul>
61 </section>
62
63 <hr>
64
65 <!-- Footer Section -->
66 <footer>
67   <p>&copy; 2026 Azmay Noor. All Rights Reserved.</p>
68 </footer>
69
70 </body>
71 </html>

```

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>My Portfolio</title>
6  </head>
7  <body>
8
9      <!-- Header Section -->
10 <header>
11     <h1>Azmay Noor</h1>
12     <p>Aspiring Web Developer and Computer Science Student</p>
13     <hr>
14 </header>
15
16 <!-- About Me Section -->
17 <section>
18     <h2>About Me</h2>
19     <p>
20         Hello! My name is Azmay Noor. I am a Computer Science student
21         with an interest in web development, programming, and technology.
22         I enjoy learning new skills, creating websites, and working on
23         projects that improve my problem-solving abilities.
24     </p>
25 </section>
26
27 <hr>
28
29 <!-- Projects Section -->

```

```

28
29 <!-- Projects Section -->
30 <section>
31     <h2>Projects</h2>
32
33     <h3>Recent Projects</h3>
34     <ul>
35         <li>
36             <strong>Personal Portfolio Website</strong> -
37             A simple HTML website showcasing my skills and projects.
38         </li>
39         <li>
40             <strong>Student Registration Form</strong> -
41             An HTML form for collecting student information.
42         </li>
43         <li>
44             <strong>Online Book Store Homepage</strong> -
45             A sample homepage designed using HTML.
46         </li>
47     </ul>
48 </section>
49
50 <hr>
51
52 <!-- Contact Section -->
53 <section>
54     <h2>Contact</h2>

```